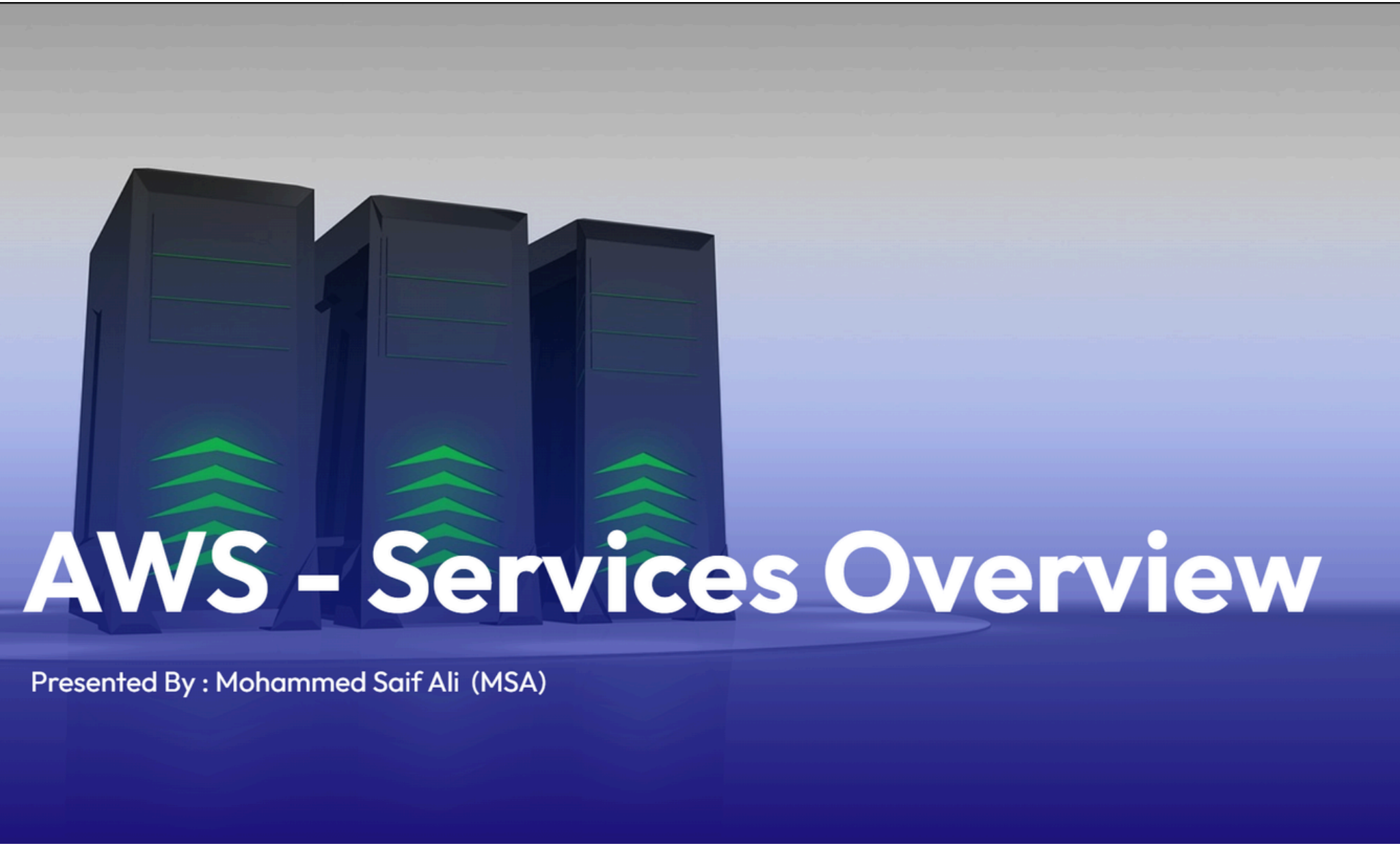


AWS - Services Overview

Presented By : Mohammed Saif Ali (MSA)

Cloud Computing

Cloud Computing is the on-demand delivery of IT resources such as servers, storage, databases, networking, and software over the internet with pay-as-you-go pricing.

Real-World Example

Instead of buying physical servers for your company, you launch EC2 instances on AWS and pay only for what you use.

Explain to a 5-Year-Old

Instead of buying your own toys, you go to a toy library and borrow toys whenever you want.

☁️ *Types of Cloud*

1.Public Cloud

Infrastructure owned by cloud providers and shared among customers.
Examples:
AWS,Azure,Google Cloud

Real-World Example

Hosting a startup website on AWS.

5-Year-Old

Like going to a public playground where everyone can play.

2.Private Cloud

Dedicated infrastructure used by only one organization.

Real-World Example

A bank running its own cloud inside its data center.

5-Year-Old

Having your own private playground at home.

3.Hybrid Cloud

Combination of Public + Private Cloud.

Real-World Example

Bank stores sensitive data on-premises but hosts the customer portal on AWS.

5-Year-Old

You play at home sometimes and at the public playground sometimes.

4.Multi-Cloud

Using multiple cloud providers together.

Real-World Example

Host application on AWS and backup data on Azure.

5-Year-Old

Using toys from two different toy stores.

Cloud Service Models

1. IaaS

Infrastructure as a Service- Provides virtual servers, storage, and networking.

Customer manages:

- OS
- Applications
- Middleware

Provider manages:

- Hardware
- Networking
- Data Center

AWS Examples:

- EC2
- EBS
- VPC

Examples:

Real World

Renting an empty apartment.

5-Year-Old

Someone gives you an empty LEGO house and you decorate it.

2. PaaS

Platform as a Service-Provides infrastructure and platform.
You only deploy code.

AWS Examples:

- EC2
- EBS
- VPC

Examples:

Real World

Renting a fully furnished apartment.

5-Year-Old

The LEGO house is already built. You just play.

3. SaaS

Software as a Service-Fully managed software over the internet.

Examples

- Gmail
- Office 365
- Zoom

Examples:

Real World

Using Gmail without managing servers.

5-Year-Old

Watching cartoons on YouTube without owning the TV station.

AWS Cloud Overview

Global Infrastructure

AWS operates data centers worldwide to provide reliable and low-latency access.

Pay-As-You-Go

Customers pay only for the resources they use without upfront costs.

Elasticity

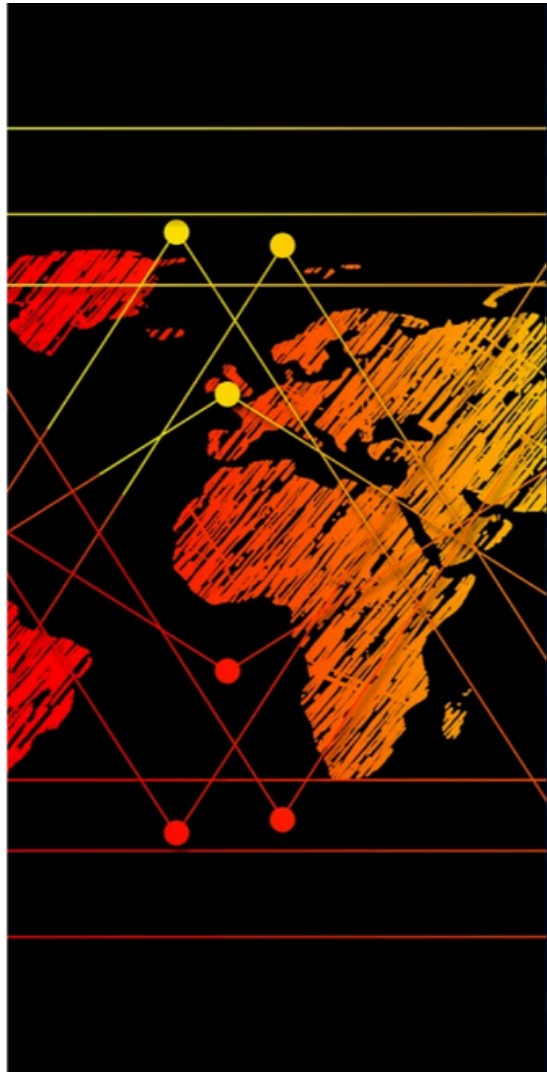
AWS scales resources automatically to meet changing demand.

Managed Services

AWS provides ready-to-use services to simplify deployment and management.

Shared Responsibility

Security responsibilities are divided between AWS and the customer.



IAM

Identity and Access Management-
Controls who can access AWS resources and what actions they can perform.

Access control in AWS IAM is the mechanism that determines who can perform which actions on what resources, under what conditions.

Examples:

Real World

Giving employees access cards to company rooms.

5-Year-Old

Mom decides which toys each child can play with.

- **IAM User-** Identity created for a human user.
  Real World - Developer login account.
  5-Year-Old- Your school ID card.

- **IAM Group-** Collection of IAM Users.
  Real World- Developers group.
  5-Year-Old- A classroom of students.

- **IAM Role**- Temporary permissions assigned to AWS services.
  Real World- EC2 accessing S3.
  5-Year-Old- Teacher allows you to borrow a toy for a while.

- **IAM Policy**- JSON document defining permissions.
  Real World- Access rules for developers.
  5-Year-Old- Rules saying what toys you can touch.

Security Foundations: IAM



Users

Individuals with their own credentials can access resources based on permissions assigned to them.



Groups

Collections of users managed together with shared permissions for easier access control.



Roles

Temporary permissions assigned to services or users to perform specific tasks securely.



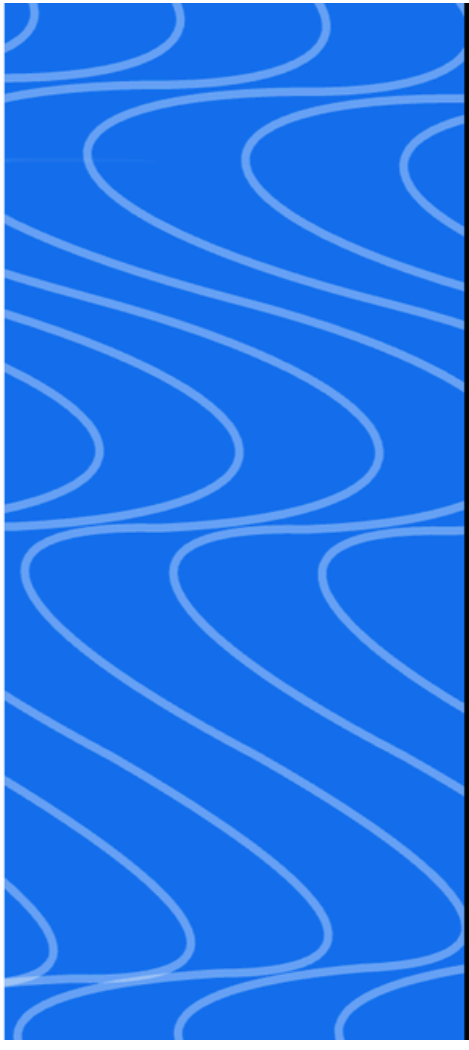
Policies

Rules defining what actions are allowed or denied for users, groups, and roles.

Compute Basics: Amazon EC2

Virtual Servers and Elastic Scaling

Amazon EC2 provides virtual servers with control over operating systems, networking, and software. Key concepts include AMIs, instance types, security groups, and Auto Scaling. A common use is hosting web servers or backend applications that scale automatically during traffic spikes.



EC2-Elastic Compute Cloud

Virtual server running in AWS.
Real World- Host applications and websites.
5-Year-Old- A computer you borrow from AWS.

EC2 Purchase Options

1. On-Demand- Pay as you go.
2. Reserved Instances- 1-Year or 3-Year commitment.
3. Spot Instances- Use unused AWS capacity. Up to 90% cheaper.
4. Dedicated Hosts- Physical server dedicated to one customer.
5. Savings Plans- Flexible discounted pricing.

EC2 Components:

1. **AMI**- Amazon Machine Image- Template containing OS and software
   Real World- Linux image with Apache pre-installed.
   5-Year-Old- Cookie cutter making identical cookies.
2. **Instance Type**-Determines CPU, RAM, storage, and networking capacity.
   Example:
   t3.micro
   5-Year-Old- Choosing a small bicycle or big bicycle.
3. **Key Pair**-Used for secure SSH login into EC2.
   Real World-SSH private/public key.
   5-Year-Old- Special key for your room.

4. **Security Group**-Virtual firewall attached to resources.
   Real World- Allow port 80 and 443.
   5-Year-Old-Security guard at a gate.

Instance Types

General Purpose-

- Web servers
- Small databases
- Applications

Compute Optimized-

- High CPU workloads
- Video encoding
- Analytics

Memory Optimized-

- SAP
- Redis
- Large databases

Storage Optimized-

- Big data
- Data warehousing

Accelerated Computing Instances-

- Machine Learning
- AI
- Graphics

Auto Scaling:

Scalability means the ability of your applications and infrastructure to handle increasing or decreasing workloads by adjusting resources automatically.

There are Two types of scalability:

Vertical scalability: Increase the size of an instance.

Horizontal scalability: Increasing the number of instances/systems.

Auto Scaling Groups:

An Auto Scaling group contains a collection of EC2 instances that are treated as a logical grouping for the purposes of automatic scaling and management.

Desired Capacity: The exact number of EC2 instances the Auto Scaling Group tries to maintain at all times.

Minimum Capacity: The lowest number of EC2 instances that must always be running in the group.

Maximum Capacity: The highest number of EC2 instances the group can scale up to.

Compute on Demand: AWS Lambda

Serverless Execution

Runs code only when triggered, eliminating the need to manage servers and enabling event-driven workflows.



Lambda Function:-

A Lambda Function is the actual piece of code that AWS executes.

It contains:

- Business Logic
- Runtime (Python, Node.js, Java, .NET, etc.)
- Environment Variables
- IAM Execution Role

Event:-

An Event is the data sent to a Lambda Function when something happens.

Events contain information about the action that occurred.

Trigger:-

A Trigger is the source that invokes (starts) the Lambda Function.

Without a trigger, Lambda does not run automatically.

Storage Choices on AWS: EBS, EFS, and S3



EBS - Block Storage

Used for single EC2 instance storage with low-latency block access. Ideal for application disks requiring high performance.



EFS - Shared File Storage

Provides a scalable file system accessible by multiple servers simultaneously. Suitable for shared files in distributed systems.



S3 - Object Storage

Offers durable, highly scalable storage for backups, media, and static website content. Designed for infrequent access and large scale.

EBS-Elastic Block Store:

Persistent block storage for EC2.

Real World- Hard disk for EC2.

5-Year-Old- Notebook attached to one child.

EFS-Elastic File System:

Shared file storage for multiple EC2s.

Real World- WordPress shared uploads.

5-Year-Old- A shared coloring book.

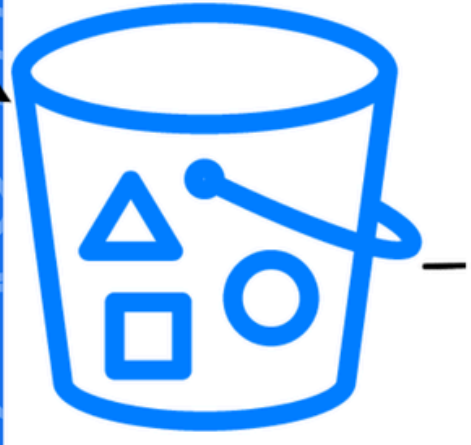
S3-Simple Storage Service:

Highly durable object storage.

Real World- Store photos, backups, videos.

5-Year-Old- Large toy warehouse.

tion to different st



S3 Infrequest Access

S3 Deep Dive: Storage Classes, Versioning, and Lifecycle Policies

01

Storage Classes

Amazon S3 offers multiple storage classes designed for different access needs and cost efficiency. This helps optimize spending by using the appropriate class for current data usage.

02

Versioning

Versioning preserves multiple variants of an object to protect against accidental deletion and enables easy recovery of previous file states.

03

Lifecycle Policies

Lifecycle policies automate moving objects between storage classes based on age or access patterns to reduce costs and manage data retention.

Amazon S3 Storage Classes

1. **S3 Standard:**
   Default storage class for frequently accessed data.
2. **S3 Standard-IA: (IA = Infrequent Access)**
   Used for data that is not accessed often but must be available immediately when needed.
3. **S3 One Zone-IA:**
   Stores data in a single Availability Zone. Cheaper than Standard-IA.
4. **S3 Intelligent-Tiering:**
   Automatically moves data between storage tiers based on access patterns.
5. **S3 Glacier Instant Retrieval:**
   Archive storage with millisecond retrieval.
   Lower storage cost than Standard.
6. **S3 Glacier Flexible Retrieval:**
   Low-cost archive storage. Retrieval takes minutes to hours
7. **S3 Glacier Deep Archive:**
   Lowest-cost S3 storage class. Retrieval can take several hours.

VPC-Virtual Private Cloud:

Amazon VPC (Virtual Private Cloud) is a logically isolated virtual network in AWS where you can launch and manage resources, similar to a traditional on premises network but with the scalability of AWS.

Subnet: *A subnet is a range of IP addresses in your VPC. You can create AWS resources, such as EC2 instances, in specific subnets.*

Route Tables: *A route table serves as the traffic controller for your virtual private cloud (VPC). Each route table contains a set of rules, called routes, that determine where network traffic from your subnet or gateway is directed.*

Internet Gateway: *An internet gateway is a horizontally scaled, redundant, and highly available VPC component that allows communication between your VPC and the internet.*

NAT gateway: *A NAT gateway is a Network Address Translation (NAT) service. You can use a NAT gateway so that instances in a private subnet can connect to services outside your VPC but external services can't initiate a connection with those instances.*

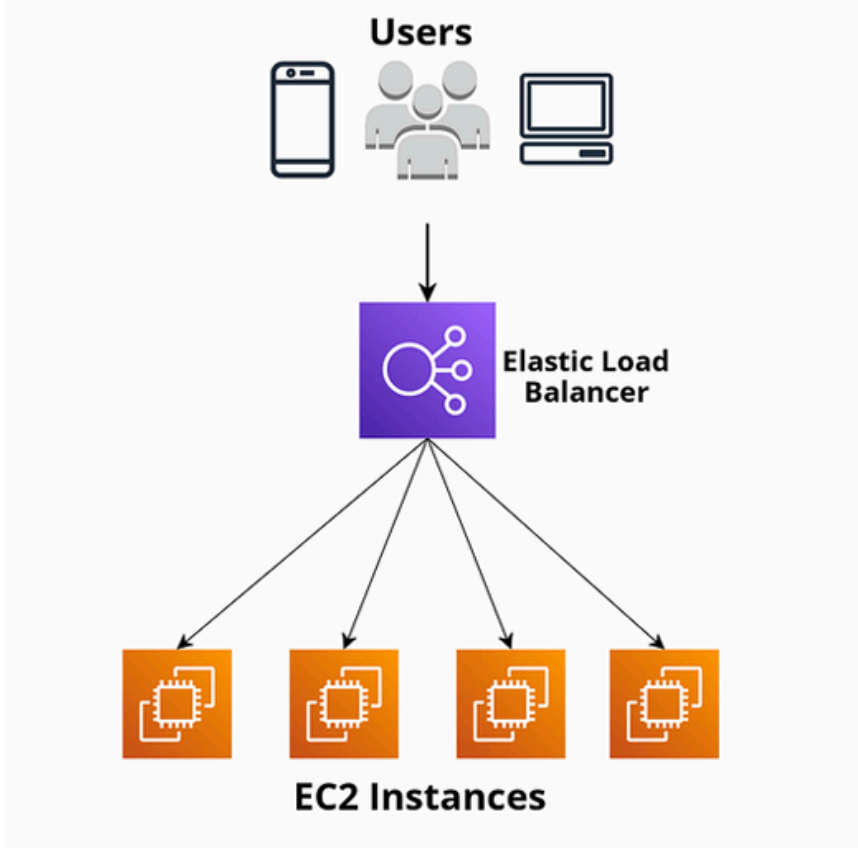
NACL (Network Access Control List):

A Network Access Control List (NACL) is a stateless firewall that controls inbound and outbound traffic at the subnet level inside a VPC.
Think of it as a security layer that protects an entire subnet.

Security Group	Network ACL
Operates at the instance (interface) level	Operates at the subnet level
Supports allow rules only	Supports allow and deny rules
Stateful	Stateless
Evaluates all rules	Processes rules in order
Applies to an instance only if associated with a group	Automatically applies to all instances in the subnets its associated with

Elastic Load Balancer:

Elastic Load Balancing automatically distributes your incoming traffic across multiple targets, such as EC2 instances, containers, and IP addresses, in one or more Availability Zones. It monitors the health of its registered targets, and routes traffic only to the healthy targets.



Application Load Balancer
Routes and load balances at the application layer (HTTP/HTTPS), and supports path-based routing. An Application Load Balancer can route requests to ports on one or more registered targets, such as EC2 instances, in your virtual private cloud (VPC).

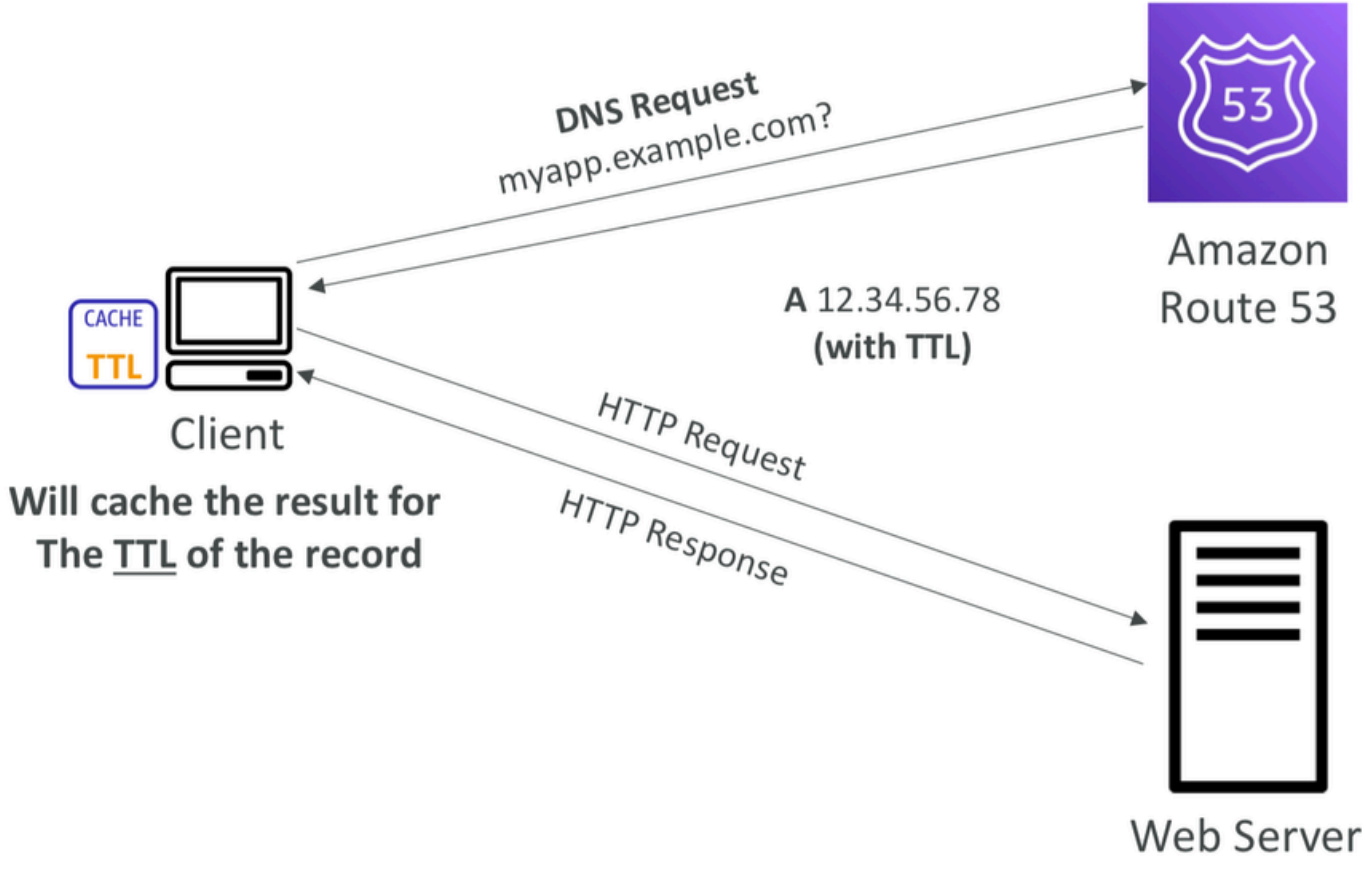
Network Load Balancer
Routes and load balances at the transport layer (TCP/UDP Layer-4), based on address information extracted from the Layer-4 header. Network Load Balancers can handle traffic bursts, retain the source IP of the client, and use a fixed IP for the life of the load balancer.

Gateway Load Balancer
Distributes traffic to a fleet of appliance instances. Provides scale, availability, and simplicity for third-party virtual appliances, such as firewalls, intrusion detection and prevention systems, and other appliances. Gateway Load Balancers work with virtual appliances that support the GENEVE protocol. Additional technical integration is required, so make sure to consult the user guide before choosing a Gateway Load Balancer.

Classic Load Balancer
Routes and load balances either at the transport layer (TCP/SSL), or at the application layer (HTTP/HTTPS).

Route 53:

Amazon Route53 is a highly available & scalable Domain Name System(DNS), web services from AWS. It translates human-readable website name into IP-address that computer use to connect.



SNS Publish-Subscribe

Delivers notifications to multiple subscribers for real-time updates and fan-out messaging.

SQS Message Queuing

Stores messages reliably until consumers process them asynchronously.

SNS, SQS, SES, and EventBridge

SES Email Delivery

Sends emails as part of application workflows for notifications and alerts.

EventBridge Event Bus

Routes application and AWS events to trigger downstream workflows based on event-driven integration.

CloudWatch, CloudTrail, and AWS Config



CloudWatch

Monitors performance with metrics and alarms to detect issues quickly.



CloudTrail

Records API activity and account actions for security investigations.



AWS Config

Tracks resource configuration changes over time for compliance reviews.

RDS and Aurora: Managed Relational Databases

Amazon RDS

- A managed relational database service that simplifies infrastructure management, backups, and patching.
- Supports Multi-AZ for high availability and Read Replicas for scaling read operations.

Amazon Aurora

- A high-performance relational database compatible with MySQL and PostgreSQL.
- Offers automated maintenance, strong scalability, and enhanced performance for growing applications.

DynamoDB



Fast, Scalable NoSQL

A fully managed NoSQL database for low-latency access and automatic scaling without server management.

Application Automation: Elastic Beanstalk and CloudFormation

01

Elastic Beanstalk Deployment

Deploy and manage applications quickly without managing infrastructure.

Launch web apps easily with minimal manual setup.

02

CloudFormation Templates

Use infrastructure as code to build and manage AWS environments.

Create consistent, repeatable deployments from templates.

CloudFormation:

AWS CloudFormation is an Infrastructure as Code (IaC) service that allows you to create, update, and manage AWS resources using YAML or JSON templates.

Template: *A Template is a YAML or JSON file that defines your AWS infrastructure. It acts as a blueprint.*
Stack: *A Stack is a collection of AWS resources created from a CloudFormation template.*
Change Sets: *Preview changes before updating a stack.*
StackSet: *Deploy the same CloudFormation stack across multiple AWS accounts and regions.*
Drift Detection: *checks if your resources were changed manually outside cloudformation.*

Elastic Beanstalk:

Elastic Beanstalk is a Platform as a Service (PaaS) that lets developers deploy applications without managing infrastructure.
You provide the application code.
AWS manages:

- *EC2*
- *Auto Scaling*
- *Load Balancer*
- *Monitoring*
- *Deployment*

Elastic Beanstalk is a PaaS service that automatically provisions and manages infrastructure required to run web applications.

Developer and Data Integration Services: Kinesis and the Code Services



Kinesis Streaming Data

Collect and process real-time streaming data such as clickstreams for analytics and insights.



Code Services CI/CD Pipeline

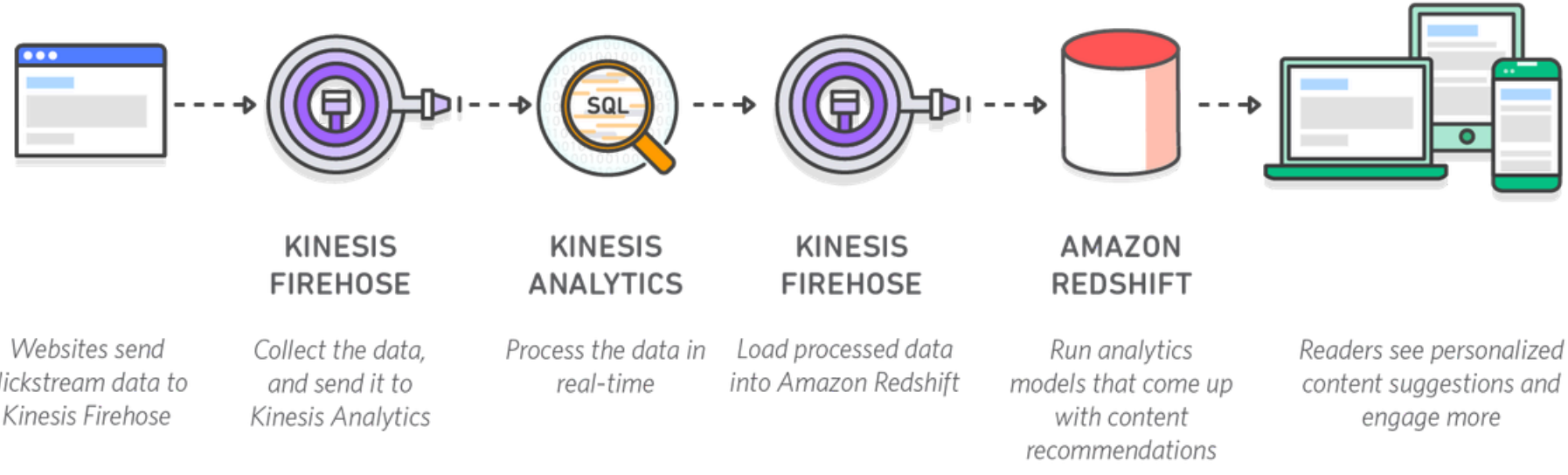
Manage source control, build, test, and deploy applications with CodeCommit, CodeBuild, CodePipeline, and CodeDeploy.

Kinesis:

Amazon Kinesis is a real-time data streaming service used to collect, process, and analyze large streams of data continuously.

Why Use Kinesis?

✅ *Real-Time Analytics*
✅ *Streaming Data*
✅ *Log Processing*
✅ *IoT*
✅ *Monitoring*
✅ *Fraud Detection*



OTHER SERVICES

Security and Governance: Secrets Manager, KMS, and Control Tower



Secrets Manager

Stores and rotates credentials securely to protect sensitive application access in sectors like healthcare and finance.



Key Management Service (KMS)

Creates and manages encryption keys ensuring data remains encrypted and protected consistently.



Control Tower

Sets up and governs secure AWS environments with centralized governance and guardrails across multiple accounts.

KMS: Encryption Keys and Data Protection



Centralized Key Management

AWS KMS enables creating and managing encryption keys to protect data at rest across services.



Integrated AWS Services

S3, EBS, RDS, and Secrets Manager use KMS-backed encryption to secure data.



Sector Applications

Finance and healthcare use KMS to keep sensitive records encrypted with strict access controls.



Data Security Importance

Encryption ensures data confidentiality and compliance with regulatory standards.

Control Tower: Multi-Account Setup and Guardrails

Centralized Governance

Provides consistent management and governance across multiple AWS accounts under one framework.

Account Vending

Automates the creation of new accounts with standardized configurations and security controls.

Enforcing Guardrails

Applies preventive and detective guardrails to ensure compliance and security policies are met.

Scalable Account Management

Supports growing organizations by standardizing and securing new accounts from day one.

Containers on AWS: ECS, ECR, EKS, and Fargate



Amazon ECS

Managed container orchestration service for running and scaling containerized applications.



Amazon ECR

Secure container image registry to store, manage, and deploy container images.



Amazon EKS

Managed Kubernetes service that simplifies running Kubernetes on AWS.



AWS Fargate

Serverless compute engine that runs containers without managing servers.

ECS - Container Orchestration

AWS's service for deploying and managing containers at scale with integrated orchestration.

ECR - Image Registry

Secure storage and management of container images used for application packaging.

ECS, ECR, EKS, and Fargate

EKS - Managed Kubernetes

A fully managed Kubernetes service that simplifies container orchestration using Kubernetes standards.

Fargate - Serverless Compute

Runs containers without managing servers, reducing operational overhead for containerized workloads.

NoSQL and In-Memory Databases: DynamoDB, ElastiCache, and MemoryDB



DynamoDB: Persistent NoSQL

Fully managed database offering fast key-value and document access with seamless scalability for modern applications.



ElastiCache: Caching Layer

In-memory cache that reduces database load and speeds up read operations to improve application responsiveness.



MemoryDB: Durable In-Memory

Durable in-memory database designed for ultra-fast workloads requiring high availability and low latency.

Analytics Storage: Amazon Redshift and Data Migration: DMS



Amazon Redshift

A data warehouse optimized for analytics and reporting on large-scale data. Redshift supports complex queries for business intelligence and dashboards.



AWS Database Migration Service (DMS)

A service that moves or replicates data between databases. Used for migrating data or ongoing replication during system transitions.

Governance and Compliance: AWS Config and AWS Organizations

01

AWS Config for Resource Tracking

Tracks configuration changes across AWS resources and evaluates compliance against policies.

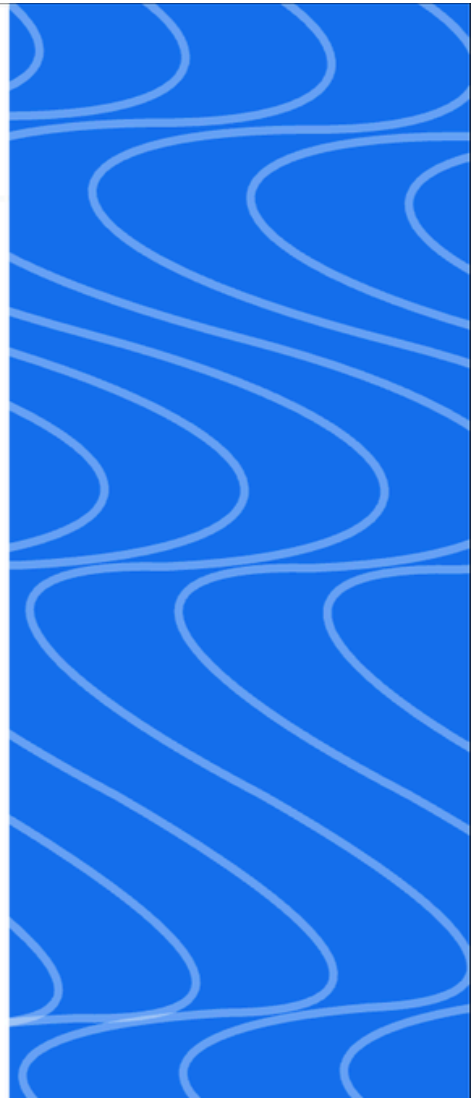
Helps maintain visibility and governance over resource states.

02

AWS Organizations for Account Management

Centrally manages multiple AWS accounts with governance and policy control.

Supports a structured environment separating development, testing, and production accounts under one management framework.



CodeCommit, CodeBuild, CodePipeline, and CodeDeploy

CodeCommit

Hosts source code repositories with version control for team collaboration.

CodeBuild

Automates code building and testing for continuous integration.

CodePipeline

Manages the release workflow from source to production deployment.

CodeDeploy

Automates deployments to compute resources for repeatable delivery.